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Studies of soot oxidation and fragmentation in a two-stage burner under fuel-lean and fuel-rich conditions

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Abstract

In this work, soot oxidation and fragmentation studies were carried out experimentally in a two-stage burner, where soot was produced in a first-stage premixed burner, while in a second stage, the soot was oxidized under fuel-lean and slightly-rich conditions. Temperature, particle size distributions (PSD), and measurements of specific gas-phase compounds were performed to evaluate the effects of temperature, O_2, and $OH^{\bullet}$ on the evolution of soot size distributions during oxidation. Results for the leanest ethylene/air flame ($\Phi_{overall} = 0.8$), showed a decrease in particle mean diameter and an increase in particle number concentration for ultrafine-sized particles with increasing height above burner (HAB), which indicates fragmentation of the fine-sized particles. At higher HAB, the soot oxidation process was dominated by soot burnout as confirmed by the decrease in number and mass concentration. The fuel rich condition ($\Phi_{overall} = 1.14$) did not show particle fragmentation; in fact, the size distribution was governed by soot burnout starting at burner surface. Measurements of gas-phase compounds, O_2, CO, CO_2, H_2, validated a detailed kinetic model, which was used to predict $OH^{\bullet}$ concentration in the secondary flame. Using the concentrations for O_2 and $OH^{\bullet}$, soot oxidation rates were calculated and compared with experimental data. Furthermore, the regions where O_2 and $OH^{\bullet}$ were dominating the process of soot oxidation were identified. Soot oxidation via O_2 was favored under fuel lean conditions close to the burner surface. Oxidation via $OH^{\bullet}$ appeared to be the major path for soot oxidation under near-stoichiometric and fuel-rich conditions after O_2 concentrations dropped, and faster soot oxidation rates were obtained.
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Keywords: Soot oxidation; Particle size distribution (PSD); Premixed flames; Soot fragmentation

1. Introduction

The process of soot formation involves several chemical and physical steps that characterize the formed carbonaceous material. It is well known that soot generated from flames undergoes pyrolysis of the parent fuel, nucleation, aromatic growth, coagulation, agglomeration and oxidation [1], where oxidation is continuously competing with

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the other processes. The oxidation process can define important characteristics such as soot size, morphology and structure. In addition, under fuel-lean conditions previous studies showed that larger particles or agglomerates may break apart and form new smaller particles [2–6]. A better understanding of the soot oxidation and fragmentation processes can help to improve the existing models for predicting size distribution and mean properties of particles as they undergo oxidation.

It is accepted that soot can be oxidized when it reacts with molecular oxygen (O_2), oxygen radical ($O^{\bullet}$), and hydroxyl radical ($OH^{\bullet}$) [1–8]. Two typical kinetic expressions to account for the soot oxidation via O_2 and $OH^{\bullet}$ were proposed by Nagle-Strickland-Constable (NSC) [7] and Neoh et al. [2,3], respectively. NSC fitted experimental data with a kinetic expression over a temperature range from 1273 to 2273 K for oxidation of pyrolytic graphite with O_2 as follows:

$$W(\mathrm{g/cm^2-s}) = \left(\frac{k_A P_{O_2}}{1 + k_z P_{O_2}}\right)x + k_B P_{O_2}(1-x) \tag{1}$$

where k_A, k_z and k_B are temperature-dependent kinetic reaction constants, P_{O_2} is the partial pressure of O_2, and x is the fraction of the surface covered with A type reaction sites. This equation has been found to fit experimental data over a narrow range of experimental conditions, particularly when graphite and O_2 are the main components in the system. Park and Appleton [9], who studied the oxidation of carbon black with O_2 in a shock tube, found higher oxidation rates for temperatures up to 4000 K compared to NSC predictions. Similar findings have been presented by other researchers in particular when high temperatures and other oxidizer species such as $OH^{\bullet}$ are present in the system [2,3,6,10].

Neoh et al. [2,3] used a two-stage burner to study the oxidation of soot over a wide range of temperatures and equivalence ratios. The results at the lowest equivalence ratio ($\Phi = 0.85$) provided data on reactivity of soot with O_2 and yielded rates slightly higher than those of NSC. The data also showed that, under fuel-lean conditions, the soot agglomerates broke up during burnout forming smaller particles. The fragmentation was reflected by the observable increase of number concentration in the nano-sized particle range, and it was attributed to the diffusion of O_2 into the soot porous network, which in turn, caused the soot agglomerates to break apart due to internal burning. This phenomenon was observed after 75–80% soot burnout [2–4]. Neoh also showed the dominant role of $OH^{\bullet}$ as a reactant on the oxidation of soot when other oxidizers species such as O_2, $O^{\bullet}$ etc. are present in the system. In Neoh's work, the rate of soot oxidation via $OH^{\bullet}$ was described by a collision efficiency Γ_{OH}, which represented the fraction of collisions of $OH^{\bullet}$ with the particle surface that resulted in the removal of a carbon atom. Neoh's kinetic expression is given by:

$$W(\mathrm{g/cm^2-s}) = 1.29 \times 10^2 \Gamma_{OH^{\bullet}} \frac{P_{OH^{\bullet}}}{\sqrt{T}} \tag{2}$$

where $P_{OH^{\bullet}}$ and T are the $OH^{\bullet}$ partial pressure and temperature. The collision efficiency was found to be 0.13 if the diameter of the individual spherical units within the aggregates is used and 0.28 if the optical (equivalent sphere) diameter is used instead. According to Neoh, the actual value of the collision efficiency should be between these two values and it is relatively independent of equivalence ratio and height within the flame. Other researchers have investigated the role of $OH^{\bullet}$ oxidation and have found similar collision efficiencies [6,10–12]. Most suggest that $OH^{\bullet}$ oxidation was the predominant reaction mechanism under near-stoichiometric and fuel-rich conditions.

The present study considers the role of O_2 and $OH^{\bullet}$ in oxidation, and takes advantage of the new developments of instrumentation to obtain particle size distributions (PSDs), temperature, morphology and structure etc. for reinvestigating the kinetics of soot oxidation via O_2 and $OH^{\bullet}$, and to consider the role of soot fragmentation with O_2.

2. Experimental section

A two-stage burner (see Fig. 1), designed after the apparatus of Neoh [2,3], was used to generate soot in a fuel-rich premixed flame, which served as the first stage. The soot was then burned in a secondary, premixed burner. In the first-stage burner, air and fuel (ethylene > 99.99 %) were added to the bottom of a 5-cm ID stainless steel chamber under $\Phi_1 = 2.5$. Complete mixing occurred over a thick bed of glass beads and the flame was stabilized over a tube bundle through which the mixture passes in laminar flow. The secondary burner and pre-oxidation chamber had multiple purposes: efficient mixing of the soot-laden combustion gases with an injection port for air to control the stoichiometry and temperature of the secondary flame; generation of a uniform flow of combustion reactants to the burner; and, oxidation of the soot from the first-stage burner. In a similar way, the secondary flame was stabilized over a tube bundle and shielded from atmospheric interference using a nitrogen shroud.

Temperature measurements, PSDs, and gas-phase measurements were carried out along the centerline of the top flame under five different fuel/air conditions ($\Phi_{overall} = 0.8$, 0.87, 0.94, 1.07 and 1.14) for atmospheric-pressure, ethylene-air flames, but only the results for 0.8 and 1.14 are reported in this paper. The overall equivalence ratio, $\Phi_{overall}$ is defined as follows:

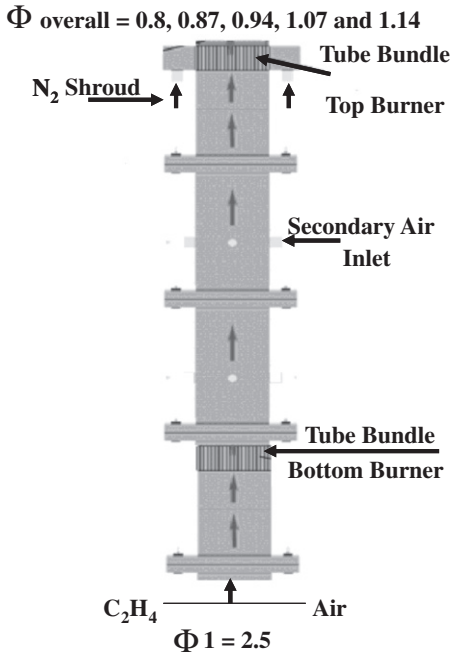

Fig. 1. Schematic of the two-stage oxidation burner.

$$\phi_{\text{overall}} = \frac{(C_2H_4/(Air_1 + Air_2))_{\text{actual}}}{(C_2H_4/Air)_{\text{stoichiometric}}} \quad (3)$$

where Air_1 and Air_2 are the amounts of air injected through the first and secondary ports respectively. The velocity of the unburned gases in the bottom burner was kept constant at 3.4 cm/sec (STP) and the secondary air was changed to reach the desired overall equivalence ratio. The total velocity (based on the total flow of air and C_2H_4) of the unburned gases for the leanest ($\Phi_{\text{overall}} = 0.8$) and richest ($\Phi_{\text{overall}} = 1.14$) conditions were 9.54 and 6.82 cm/s (STP), respectively.

Temperature profiles were measured using a 0.02032 cm diameter, type-B thermocouple. Experimental setup and radiation corrections were similar to that of Rosner [13]. PSDs were measured with a scanning mobility particle sizer (SMPS), which consisted of a TSI 3080 classifier with a Model 3085 Nano DMA and a TSI 3025 ultrafine condensation particle counter (UCPC). Two different impactor sizes (0.071 and 0.0457 cm) were used to measure PSDs in the range from 3 to 160 nm. The SMPS sampling system was similar to that of Zhao et al. [14] with some minor changes in probe size (OD = 1.1cm) and pinhole diameter (0.024 cm). Dilution ratios were in the range of 10^3 to 10^4 which minimized particles losses and coagulation in the sampling system. Corrections for diffusion losses and coagulation along the sampling line were carried out following the procedure presented by Minutolo et al. [15]. Corrections due to diffusion losses in the SMPS were conducted using the AIM software. H_2, O_2, CO and CO_2 were sampled using a water-cooled probe (0.2-cm ID) and analyzed by online gas chromatography (VARIAN, CP-4900 Micro GC).

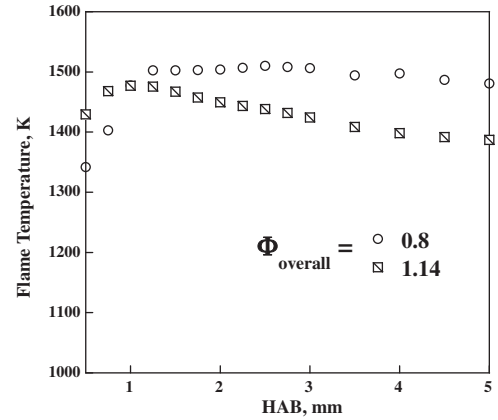

Fig. 2. Top burner temperature profiles as a function of HAB for Φoverall = 0.8 and 1.14.

3. Results and discussion

3.1. Temperature profiles and evolution of the soot size distribution

Temperature profiles were measured from the second burner surface up to 5 mm. Figure 2 illustrates the results for $\Phi_{\text{overall}} = 0.8$ and 1.14. The difference in peak temperature is consistent with the evolution of the peak temperature in premixed flames as a function of the equivalence ratio. These temperature profiles were used as inputs into a detailed kinetic model to predict $OH^{\bullet}$ radical concentration profiles (see Section 3.2).

Soot size distributions were obtained as a function of the HAB for the lean and rich flames studied. Figure 3 presents the results of PSDs for $\Phi_{\text{overall}} = 0.8$ (left) and $\Phi_{\text{overall}} = 1.14$ (right) conditions in the locations where major changes took place. Data for the leanest flame ($\Phi_{\text{overall}} = 0.8$) at HAB = 0 mm (Fig. 3, left), showed that the initial distribution was mostly in the fine mode (particles diameter > 10 nm) with little contribution of particles less than 10 nm. This PSD did not change significantly up to 2.5 mm where particles in the ultrafine mode (particles diameter $\leqslant$ 10 nm) started to appear. In the range of HAB from 2.5 to 3 mm, particle mean diameter decreased from 36.7 to 3.7 nm, and at higher HAB, the particles started to burnout significantly, evidenced by a drop in number concentration. For $\Phi_{\text{overall}} = 1.14$ (Fig. 3, right), particles in the ultrafine mode almost disappeared when they reached HAB = 2 mm, and they are difficult to see at higher elevations. For

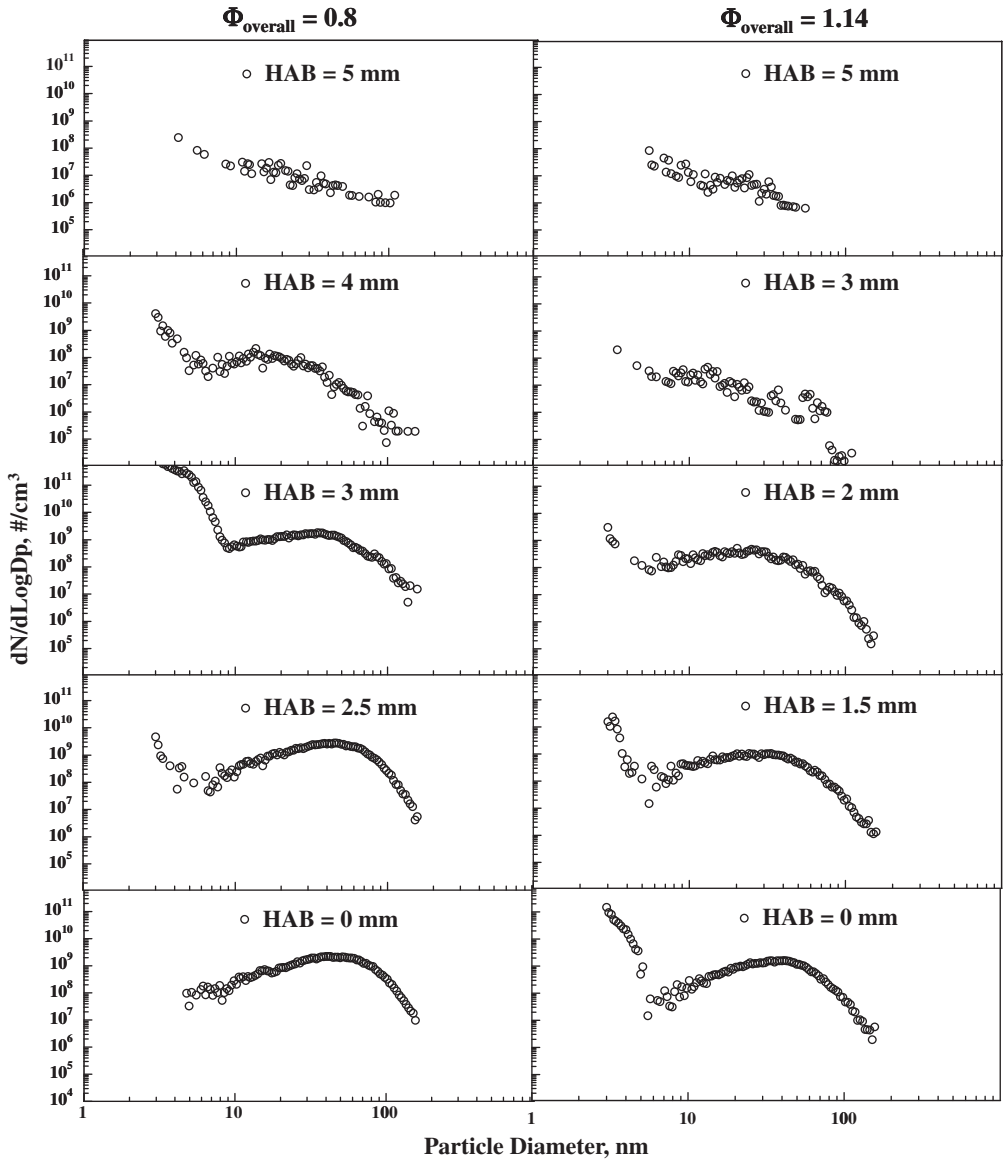

Fig. 3. Experimental PSDs measured as a function of the height above the top burner for $\Phi_{overall} = 0.8$ (left) and $\Phi_{overall} = 1.14$ (right).

the fine mode, the rate of oxidation increased significantly after HAB = 2 mm, and soot burnout dominated the oxidation process.

Particles in size intervals of less than and greater than 10 nm (the detection limit of the SMPS was 3 nm) were integrated to provide measures of the number and mass concentration of the ultrafine and fine modes (Fig. 4a and b, for the lean flame and Fig. 4c and d for the rich flame). For the lean flame, the particle number (Fig. 4a) and mass concentration (Fig. 4b) in the ultrafine mode were observed to increase up to HAB = 3 mm. This behavior has been attributed to oxygen diffusion into the pores of the soot particles causing internal burning, which, in turn caused the soot aggregates to break apart [2,3,10]. It is also interesting to notice that this fragmentation took place for soot burnout less than 5%, which is considerably lower than the value observed by Neoh and coworkers (80% burnout). At HAB > 3 mm, particles in the ultrafine mode were almost completely combusted, as evidenced by the decrease in number and mass concentration. For the fine mode (Fig 4a and b), mass and number remained relatively constant up to 2.5 mm, and then started to decrease in the region between 2.5 to 3 mm; this can be attributed

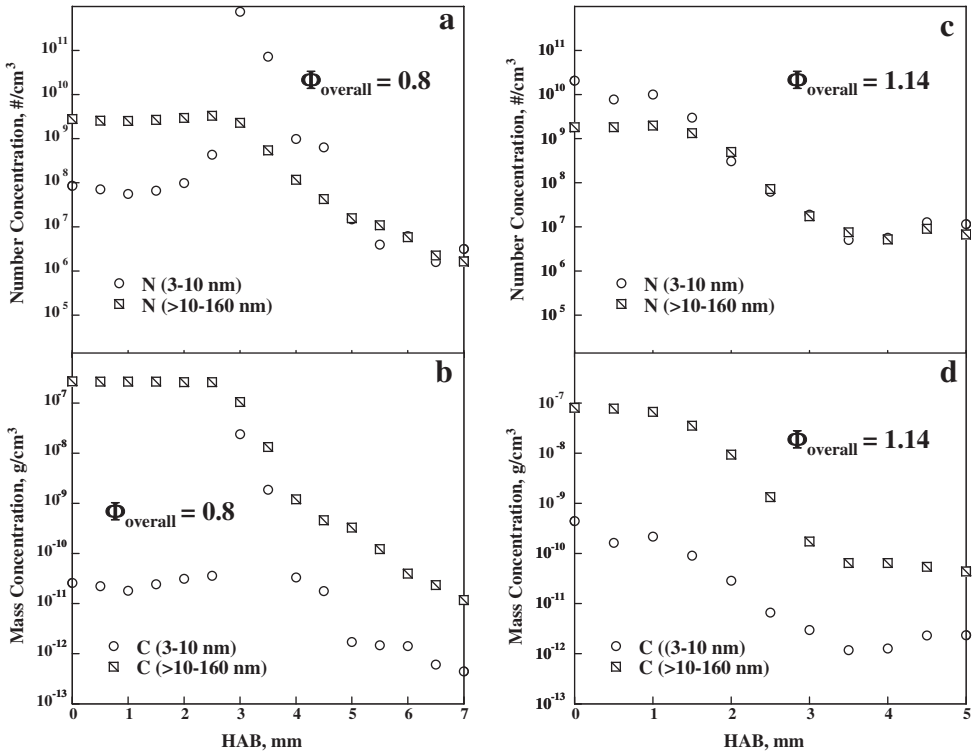

Fig. 4. Integrated number and mass concentrations for fuel lean and fuel rich flames. (a) Number concentration as a function of HAB for the ultrafine and fine modes ($\Phi_{overall} = 0.8$) (b) Mass concentration as a function of HAB for the ultrafine and fine modes ($\Phi_{overall} = 0.8$) (c) Number concentration as a function of HAB for the ultrafine and fine modes ($\Phi_{overall} = 1.14$) (d) Mass concentration as a function of HAB for the ultrafine and fine modes ($\Phi_{overall} = 1.14$).

to the fragmentation and oxidation of the bigger particles. In the upper region of the flame, soot was almost completely burned out. For the richest flame (Fig. 4c and d), both ultrafine and fine modes showed a decrease in number and mass concentration up to 3 mm. Higher in the burner, they leveled off with only minor changes in number or mass concentration. No evidence of fragmentation was observed for this flame in the range of HAB from 0 to 5 mm.

3.2. Concentrations of H_2, O_2, CO, CO_2 and $OH^{\bullet}$

The concentration profiles of major gas species, H_2, O_2, CO and CO_2, were measured experimentally within the secondary flame using an isokinetic sampling system coupled to a gas-chromatograph. Experimental temperature profiles (Fig. 2), and experimental conditions measured at the surface of the top burner (mass flow rate, H_2, CO, CO_2, O_2 concentration, and PSDs) were used as inputs into a detailed kinetic model to predict the $OH^{\bullet}$ concentrations as a function of the HAB. The model, developed by D'Anna and coworkers [16], couples gas-phase chemistry with a sectional approach to follow the transition from gas-phase to nascent particles and soot oxidation. The details of the model have been reported previously [16,17]. Although the model does not include reactions pathways that account for the fragmentation of particles, its gas-phase chemistry has been used in previous investigations to predict the concentration profiles of radicals such as $H^{\bullet}$, $O^{\bullet}$, $OH^{\bullet}$ etc. in premixed flames.

Figure 5a and b compare the experimental measurements (open symbols) of H_2, O_2, CO and CO_2 to model predictions (solid and dashed line) for $\Phi_{overall}$ of 0.8 and 1.14 respectively. Predictions of $OH^{\bullet}$ are also presented and they are plotted with the right-side scale. Good agreement was obtained between the model results and the experimental data for these flames. Sampling effects due to the water-cooled probe were taking into account by shifting the model downstream as supported in previous investigations [14,17]. As seen in these figures, at the burner surface (HAB = 0 mm), the main fuel components for the secondary, premixed flame were H_2 and CO. In the leanest flame, H_2 and CO were consumed in the upper region of the flame, oxygen dropped to about 6%, and CO_2 concentrations increased and leveled. According to predictions, $OH^{\bullet}$ started to increase in the region where O_2 decreased and reached a maximum when the con-

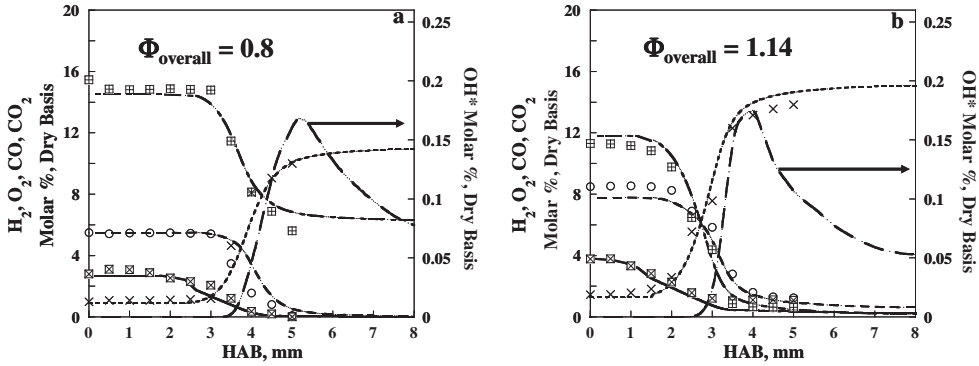

Fig. 5. Experimental measurements (symbols) versus model predictions (lines) of H_2 (⊠, data, — model), O_2 (⊞ data, -·— model), CO (O data, – – – model), CO_2 (X data, - - - - model) and $OH^{\bullet}$ (.- — model) for (a) $\Phi_{overall} = 0.8$ and (b) $\Phi_{overall} = 1.14$.

centration of O_2 leveled off for HAB > 4 mm. The concentration of O_2, CO and CO_2 started to show a significant change after an induction period up to 2.5–3 mm which is consistent with the trend of the PSDs obtained in the second burner, where major changes in number and mass concentration occurred. On the other hand, H_2, CO, O_2, CO_2 and $OH^{\bullet}$ were consumed and formed closer to the burner surface for the rich condition; the higher $OH^{\bullet}$ concentrations favor faster soot oxidation resulting in little or no fragmentation of the particles in this flame. The effect of other radicals and species, such as $O^{\bullet}$, H_2O, CO_2, and CO etc. on soot oxidation were not considered since previous studies [2,3,8] have shown that these species contributed negligibly to the oxidation under experimental conditions similar to those in the present experiments.

3.3. Experimental and predicted soot oxidation rates

Soot oxidation rates (W) were calculated from the experimental PSDs as a function of the height above the second burner from:

$$W(\mathrm{g/cm^2 - s}) = \frac{1}{A}\frac{dm}{dt} \tag{4}$$

where A was the surface area of soot particles per unit volume of gas, m was the total soot mass per unit volume of gas, and t was time. Area and mass were calculated directly from the SMPS size and number concentration assuming spherical particles. Allowance for the change of soot density with mobility diameter followed the procedures of Maricq et al. [18]. The time interval was obtained from measurements of temperature, cross sectional area and volumetric flow rate. The results for the experimental W calculated using Eq. (4) are presented in Fig. 6a for $\Phi_{overall} = 0.8$ and 1.14. For $\Phi_{overall} = 0.8$, rates were found to be relatively low and constant close to the burner surface, increasing in the upper (HAB > 2.5 mm) region of the flame. For the rich flame ($\Phi_{overall} = 1.14$), rates increased faster close to the burner surface and reached a maximum of 7.8×10^{-4} g/cm^2-s before starting to decrease. This behavior seems to be consistent with the trend of the PSDs (Fig. 3), and O_2 and $OH^{\bullet}$(see Fig. 5b) concentrations profiles where small changes took place up to 1.5–2 mm.

Figure 6b shows the ratio (R) of oxidation rates predicted using NSC (Eq. (1)) and Neoh's (Eq. (2)) equations. The figure illustrates regions where either soot oxidation via O_2 or via $OH^{\bullet}$ are dominant in the two stage burner. For $\Phi_{overall} = 0.8$, we can see that the oxidation of soot via O_2 might be predominant at HAB < 3 mm since R is increasing, in particular in the region where O_2 was decreasing and fragmentation of particles took place. At higher elevations (HAB > 3 mm), the ratio R decreases indicating a major increase in oxidation via $OH^{\bullet}$ which was consistent with the higher rates obtained from PSD data. For $\Phi_{overall} = 1.14$ a similar behavior was observed; however, major changes occurred closer to the burner surface. Due to probe effects, uncertainty in the actual HAB was ~±1 mm. The greater depths of O_2 penetration, because of its lower oxidation rate relative to $OH^{\bullet}$, may be responsible for the fragmentation of the particles seen for $\Phi_{overall} = 0.8$. In both flames, $OH^{\bullet}$ oxidation rates were faster than O_2, and maximum oxidation rates calculated using Neoh et al.'s collision efficiency of 0.13 for $OH^{\bullet}$ were on approximately 80 % of the maximum experimental value, the difference between measured and calculated values being well within the range of uncertainty of values reported in the literature for the collision efficiency.

4. Conclusions

Soot oxidation and fragmentation processes were studied in a two stage burner by measuring

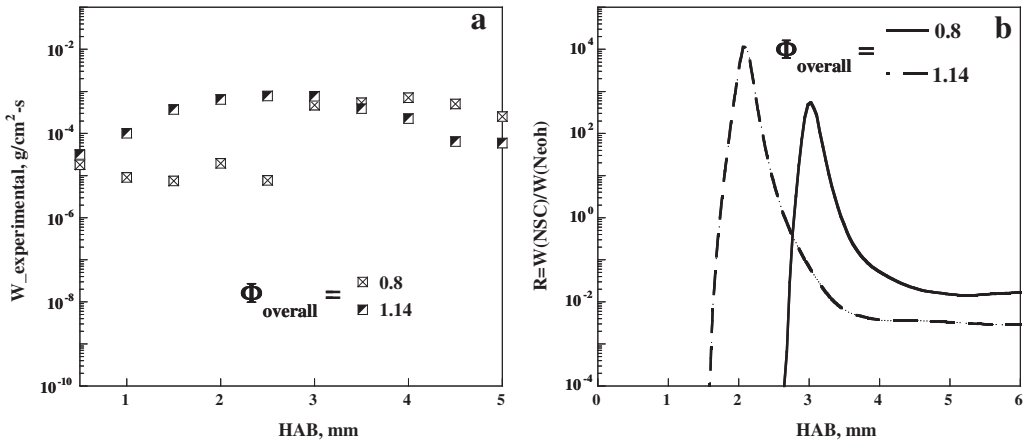

Fig. 6. (a) W calculated from the evolution of PSD and (b) Ratio between W predicted using NSC and Neoh's models as a function of the HAB for $\Phi_{overall} = 0.8$ and 1.14.

changes in particle size and concentration as a function of the HAB under fuel lean and rich conditions. Comparisons were performed for $\Phi_{overall}$ 0.8 and 1.14. The most remarkable difference found between $\Phi_{overall} = 0.8$ and 1.14 was the observable increase in number concentration in the ultrafine mode for the lean flame as compared to the rapid disappearance of nano-particles in the rich flame. This result for the lean flame suggests fragmentation of the soot aggregates due to O_2 penetration leading to breakup of the bridges cementing the primary particles in the aggregates. By contrast, no evidence of fragmentation was present for the rich flame, and the process was dominated by soot burnout over the entire HAB. Analysis of the results in terms of gas-phase species, PSDs, and soot oxidation rates provided information about the contributions of O_2 and $OH^{\bullet}$ on soot oxidation and fragmentation. For $\Phi_{overall} = 0.8$, O_2 oxidation was predominant close to burner surface, and favored the fragmentation of particles. Downstream, $OH^{\bullet}$ oxidation became predominant, and soot oxidation rates increased. At $\Phi_{overall} = 1.14$, soot oxidation rates were higher close to burner surface which can be attributed to the faster consumption of O_2 and earlier formation of $OH^{\bullet}$.

A major difference between the results in this paper and those of Neoh et al. is that the fragmentation occurred at low soot burnouts for this paper and at high conversions for Neoh et al. This is believed to be consistent with percolation theory which predicts fragmentation at high burnout percentages, approximately 80% varying with pore structure. For the present work, it appears as though the percolative fragmentation involved the bridges between primary particles and occurred at low overall soot burnout; in contrast, for Neoh et al. the fragmentation occurred at high soot burnout and involved the break up of primary particles.

Acknowledgments

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